

Analysis General Exam - August 2011

Closed book, closed notes. Please pledge. In each problem, justify all assertions, show calculations, and identify those theorems which you invoke in your arguments.

Problem 1

(a) Suppose that f is analytic in an open set containing the closed disk $\{z : |z| \leq R\}$ and that a, b are two complex numbers with $|a| < R$ and $|b| < R$. Evaluate

$$\int_{\gamma_R} \frac{f(z)}{(z-a)(z-b)} dz$$

where γ_R is the positively oriented circle centered at 0 with radius R .

(b) Using your work in (a), prove Liouville's theorem on bounded entire functions.

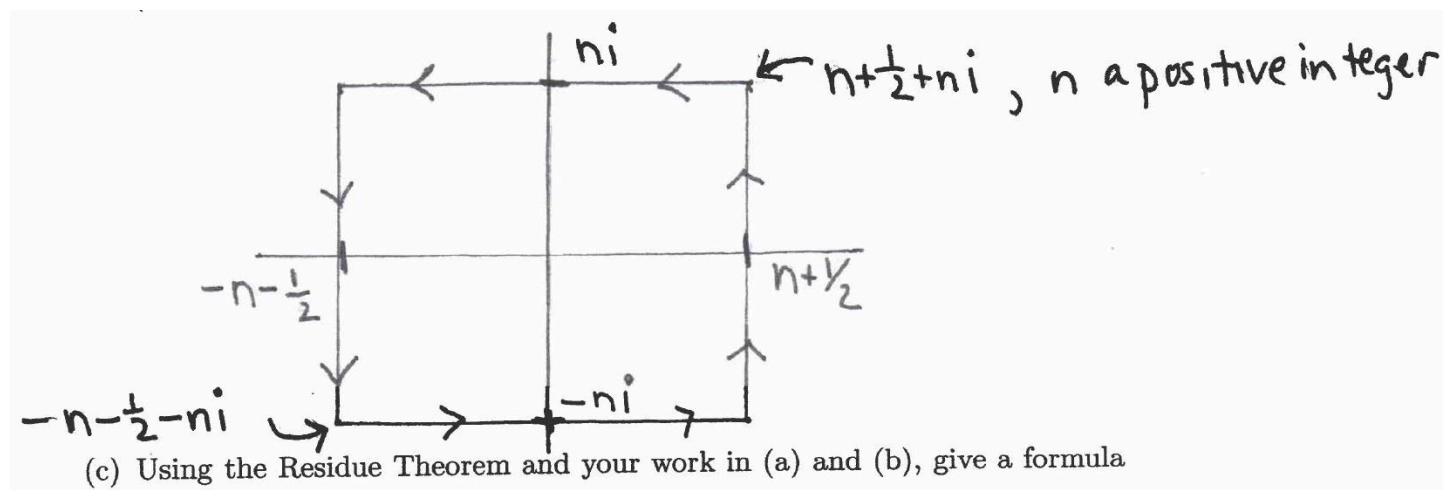
Problem 2

Let a be a real number that is not an integer and let

$$f(z) = \frac{\pi \cos \pi z}{(z+a)^2 \sin \pi z}$$

(a) Compute the residue of f at each of its singularities.

(b) Consider the rectangle γ_n as shown. Show that $\int_{\gamma_n} f(z) dz \rightarrow 0$ as $n \rightarrow \infty$. (You may give "order of magnitude" estimates in doing this.)



The rectangular contour has real coordinates from $-n - 1/2$ to $n + 1/2$ and imaginary coordinates from $-n$ to n , where n is a positive integer. Its upper-right corner is $n + 1/2 + ni$ and its lower-left corner is $-n - 1/2 - ni$. Arrows run rightward along the bottom, upward along the right side, leftward along the top, and downward along the left side: counterclockwise orientation.

(c) Use your work in (a) and (b) to find a formula for

$$\sum_{k=-\infty}^{\infty} \frac{1}{(k+a)^2}$$

(d) What is

$$\sum_{k=-\infty}^{\infty} \frac{1}{(2k+1)^2}?$$

Problem 3

Suppose that $a_1, a_2, \dots, a_n$ are n points in the open unit disk $\mathbb{D} = \{z : |z| < 1\}$. Set

$$F(z) = \prod_{m=1}^n \frac{z - a_m}{1 - \overline{a_m}z}$$

Show that for each $c \in \mathbb{D}$, the equation

$$F(z) = c$$

has n roots in $\mathbb{D}$ (counting multiplicities).

Problem 4

A function $f(z)$ that is analytic in the unit disk $\mathbb{D} = \{z : |z| < 1\}$ is said to be subordinate to the analytic function $F(z)$ if $f(z) = F(g(z))$ for some function $g(z)$ that is analytic in $\mathbb{D}$ and satisfies $|g(z)| \leq |z|$ there. This is written $f \prec F$.

(a) Show that if $f \prec F$, then $|f'(0)| \leq |F'(0)|$.

(b) Suppose that f is analytic in $\mathbb{D}$, $f(0) = 0$ and $|\operatorname{Re} f(z)| < 1$ for all $z \in \mathbb{D}$. Set

$$F(z) = \frac{2}{\pi} i \log \frac{1+z}{1-z}$$

Show that $f \prec F$. (Log denotes the principal branch.)

(c) Let f be as in (b). Show that $|f'(0)| \leq \frac{4}{\pi}$.

Problem 5

Let (X, Σ, μ) be a measure space, Σ a σ -algebra of sets, μ a measure defined on Σ . Let $\mathcal{H} = L^2(X, \Sigma, \mu)$ be the Hilbert space of square-integrable Σ -measurable functions. Let $\Sigma_0 \subset \Sigma$ be a sub- σ -algebra of Σ and let $\mathcal{H}_0 = L^2(X, \Sigma_0, \mu)$ be the subspace of $\mathcal{H}$ consisting of functions in $\mathcal{H}$ which are Σ_0 -measurable.

Let f be a function in $\mathcal{H}$. Show that there is a function f_0 in $\mathcal{H}_0$ (in particular Σ_0 -measurable) such that

$$\int_X f_0(x)g(x)d\mu(x) = \int_X f(x)g(x)d\mu(x) \text{ for all } g \in \mathcal{H}_0$$

Explain your reasoning.

Problem 6

(a) Let $F(x, y)$ be a continuous real-valued function defined on the closed square $[0, 1] \times [0, 1] \subset \mathbb{R}^2$. Show that

$$g(x) \equiv \sup_{y \in [0, 1]} F(x, y)$$

is lower semi-continuous in the sense that for each $x \in [0, 1]$ and $\epsilon > 0$, there is a δ such that

$$g(z) > g(x) - \epsilon \text{ for } |z - x| < \delta$$

(b) Let $f(x)$ be a continuous real-valued function defined on $[0, 1]$, and for $\kappa > 0$ define

$$A_\kappa \equiv \{x \in [0, 1] : |f(x) - f(y)| \leq \kappa|x - y| \text{ for all } y \in [0, 1]\}$$

By considering $F(x, y) \equiv |f(x) - f(y)| - \kappa|x - y|$, show that the complement A_κ^c of A_κ is open, hence A_κ is closed.

(c) Show that for $\kappa > 0$,

$$B_\kappa \equiv \{x \in [0, 1] : |f(y) - f(x)| < \kappa|x - y| \text{ for all } y \in [0, 1], y \neq x\}$$

is a Borel measurable set of the real line.

Problem 7

Consider the real-valued function $F(x)$, $x \in \mathbb{R}$, defined by the (improper) Riemann integral

$$F(x) = \int_0^\infty \frac{\cos(xt)dt}{1+t}$$

Show that $F(x)$ is a continuous function of x for $0 < x < \infty$. Hint: First integrate by parts to obtain an absolutely convergent integral.

Problem 8

Let $\{a_n\}_{n=1,2,\dots}$ be a square summable sequence of complex numbers such that $\sum_n |a_n|^2 = 1$, and set

$$f_r(x) = \sum_{n \geq 1} r^n a_n e^{inx}$$

for $0 \leq r < 1$.

- (a) Show that for each $r \in [0, 1)$, the series defining $f_r(x)$ converges for each x , and that $f_r(x)$ is a bounded function of x . In particular, $f_r(x)$ is in $L^2([0, 2\pi], dx)$, the space of square-integrable functions on the interval $[0, 2\pi]$ with Lebesgue measure.
- (b) Let $\{r_j\}_{j=1,2,\dots}$ be a sequence of numbers in $[0, 1)$ such that $\lim_{j \rightarrow \infty} r_j = 1$. Show that the sequence of functions $\{f_{r_j}\}$ is an L^2 -Cauchy sequence.
- (c) Let

$$f = \lim_{j \rightarrow \infty} f_{r_j}$$

the limit meaning in an L^2 -sense. Show that

$$a_m = \frac{1}{2\pi} \int_{[0, 2\pi]} e^{-imx} f(x) dx$$

for all integers m .